

Mood-Based Wellness Recommendation System

A Bring-Your-Own-Project (BYOP) Report

Name: Ananya Parmar

Registration Number: 25BAI10960

Branch: B.Tech CSE (AI/ML)

Year: First Year

Course Code: CSA2001

Institution: VIT Bhopal University

INTRODUCTION

Mental and emotional well-being is an important part of our lives, particularly for students and youth who have to deal with various challenges, uncertainties, and mood swings. Music, activities, and self-care practices have been known to affect mental health in a positive manner. However, people face difficulties in choosing the kind of music or activities they should choose to match their mood.

The Mood-Based Wellness Recommendation System is a simple application developed using Python, where a user can input their mood and be provided with suitable music and wellness activities to match their mood. This project aims to incorporate basic data handling skills with user interaction to create a meaningful application.

Problem Statement

Many people have mood swings, but there are no immediate resources available for these users. The available resources are either too complex due to the availability of too many options or not related to the user's mood.

There is a need for a simple system that:

- Accepts the user's mood as input
- Provides suggestions for music genres
- Provides suggestions for wellness activities
- Does not require internet access

This project has tried to fulfill this need by utilizing Python programming.

Objectives

The main objectives of this project are:

- Designing a mood-based recommendation system using Python
- Mapping the mood with the right type of music and wellness activities
- Creating a simple interactive user interface
- Applying the concepts of data handling, conditional statements, and modular programming
- Creating a project that can positively impact the emotional well-being of the user

Scope of the Project

The scope of this project includes:

- Mood selection through a user interface
- Recommendation of music genres (not actual audio streaming)
- Suggestion of wellness activities such as journaling, walking, meditation, etc.
- Offline execution using local files

The project does not include:

- Machine learning or AI prediction
- Real-time music streaming
- User authentication or data storage

Tools and Technologies Used

- Programming Language Used: Python
- Libraries Used:
 - pandas – for handling the dataset
 - tkinter/streamlit – for the UI (depending on the implementation)
- Dataset Format:
- Environment Used for Development:
- Operating System Used for Development: Windows

Dataset Description

The dataset used in this project is a CSV file containing mood-based information. Each record consists of:

- Mood – Emotional state such as Happy, Sad, Anxious, Calm, Energetic
- Music Genre – Recommended genre suitable for the mood
- Wellness Activity – Suggested activity to improve or stabilize mood

Example Entries:

- Mood: Happy → Music: Pop → Activity: Dancing or creative work
- Mood: Sad → Music: Soft acoustic → Activity: Journaling or listening quietly
- Mood: Anxious → Music: Lo-fi → Activity: Breathing exercises

This dataset allows easy filtering and recommendation generation.

System Architecture

The system follows a simple architecture:

1. User selects their current mood
2. The application reads the dataset
3. Matching records are filtered
4. Music and activity recommendations are displayed

The logic is modular, ensuring clarity and maintainability.

Methodology

The project was developed using the following steps:

1. **Requirement Analysis**
Identified user needs and defined project goals.
2. **Dataset Creation**
Created a CSV file mapping moods to music genres and wellness activities.
3. **Backend Logic**
Implemented mood selection and filtering using pandas.
4. **User Interface Design**
Built an interactive interface to allow users to select moods and view recommendations.
5. **Testing**
Tested the system for different moods and edge cases

Implementation Details

The application reads the dataset using pandas and stores it in a DataFrame. When a user selects a mood, the system filters rows matching that mood. Recommendations are displayed dynamically.

Key features include:

- Dropdown menu for mood selection
- Dynamic recommendation display
- Error handling for invalid inputs

The modular design ensures that additional moods or activities can be added easily.

Challenges Faced

Some challenges encountered during development include:

- Handling dataset column mismatches
- Managing user input validation
- Designing a clean and readable interface
- Ensuring recommendations are meaningful and balanced

These challenges were resolved through debugging, restructuring the dataset, and simplifying logic.

Results and Output

The system successfully:

- Accepts user mood input
- Displays appropriate music genres
- Suggests helpful wellness activities

The output is instant, easy to understand, and user-friendly.

Applications

This system can be used:

- By students for emotional regulation
- As a base for mental wellness apps
- For learning recommendation system basics
- As a prototype for AI-based wellness platforms

Future Enhancements

Possible future improvements include:

- Adding machine learning for personalized recommendations
- Integrating music APIs (Spotify, YouTube)
- Mood detection through text or voice input
- User profiles and recommendation history
- Mobile application version

Conclusion

The Mood-Based Wellness Recommendation System demonstrates how simple programming techniques can be used to create meaningful applications. By combining mood awareness with music and wellness activities, the project promotes emotional balance and self-care. The project fulfills its objectives and provides a strong foundation for future expansion.

References

1. Python Documentation
2. pandas Official Documentation
3. Mental Health and Music Research Articles
4. Online wellness resources